

VAIBHAV KARBHANTNAL

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SUMMARY

Resourceful MCA student with a proven ability to build and deploy end-to-end applications. Combines a strong foundation in machine learning with practical backend development skills in Python, Flask, and FastAPI. Passionate about leveraging data to create robust, scalable, and impactful software solutions.

SKILLS

Languages	Python, C++, Java
Data Engineering	Apache Spark, AWS
API	RestAPI, FastAPI
Databases	SQL (MySQL), NoSQL (MongoDB)
Containerization	Docker
Data Science	Machine Learning, Deep Learning, Data Analytics
Web Development	HTML, CSS, JavaScript, Node js

INTERNSHIP

Backend Developer Intern

RoborosX Omni Tech Solutions LLP

May 2025 - June 2025

Bangalore, Karnataka

Skills/Tools used: Python, Flask, MySQL, REST API, Postman, Git/GitHub, Render

Contribution:

- Designed, developed, and deployed a production-ready Internal Mailing System using Python and Flask to centralize communication across internal services.
- Built and tested robust RESTful APIs for user authentication, mail dispatch, and message handling, ensuring system reliability and functionality.
- Engineered an innovative image-based tracking mechanism to provide email read receipts.
- Successfully deployed the system to a cloud platform (Render), leading to its integration and active use within the company's production environment.

Subject Matter Expert (SME)

Chegg (Direction Educare)

June 2023 – November 2023

Punjab, India

Skills/Tools used: Microsoft Word, Excel, PowerPoint and subject-specific academic knowledge.

Contribution:

- Acted as a Subject Matter Expert (SME) to provide detailed insights and develop comprehensive solutions to complex academic problems, enhancing student learning experiences.
- Conducted in-depth technical analysis to ensure the accuracy and relevance of educational content.
- Collaborated with other SMEs and teams through regular Zoom calls to ensure high-quality content delivery and effective problem-solving.

Full Stack Web Development Intern

RoborosX Omni Tech Solution LLP

May 2023 - September 2023

Bangalore, Karnataka

Skills/Tools used: HTML, CSS, Node.js, REST API, Git/GitHub, AJAX, Bootstrap

Contribution:

- Collaborated with cross-functional teams to design, develop, and deploy web solutions.
- Attended seminars focused on the latest web development trends and technologies.
- Gained hands-on experience with both front-end (HTML, CSS, AJAX, Bootstrap) and back-end (Node.js, REST API).

PROJECTS

Real-Time Forest Fire Risk Prediction

MCA Final Sem Project

A full-stack web application to monitor, predict, and visualize forest fire risk levels across India in real-time using live NASA satellite data.

- Developed a Python backend (FastAPI/Flask) to fetch and preprocess data from the NASA FIRMS API.
- Integrated a pre-trained machine learning model to classify fire risk as Low, Moderate, or High.
- Built a dynamic frontend with Streamlit and Folium to display interactive maps with features like heatmaps, marker clusters, and user-driven filters.

Tech Stack: Python, FastAPI, Streamlit, Pandas, Scikit-learn, Folium, NASA FIRMS API.

Internal Mailer System

Solo Project (Internship)

A secure and scalable backend system designed to manage all internal email communications, which was successfully deployed into the company’s production environment.

- Built a RESTful API using Flask to handle email sending, scheduling, tracking, and template management.
- Designed and managed a MySQL database for storing user data, email logs, and scheduled tasks.
- Implemented role-based access control, attachment validation, and an email-open tracking system.

Tech Stack: Python, Flask, MySQL, SQLAlchemy, APScheduler, Jinja2.

Customer Segmentation using ML

MCA 2nd Sem Project

A data-driven project to segment customers into distinct groups, enabling targeted marketing strategies and improved customer engagement.

- Performed data cleaning, preprocessing, and feature engineering on a customer dataset.
- Applied K-Means clustering to identify initial segments and utilized an XGBoost model to refine classification accuracy.
- Employed GridSearchCV for hyperparameter tuning to optimize the model’s predictive performance.

Tech Stack: Python, Scikit-learn, Pandas, XGBoost, Jupyter Notebook.

ACHIEVEMENTS & INTERESTS

- Awarded **Student of the Year 2023** in the BCA program for academic excellence and significant contributions to university projects and community initiatives.
- Interests include:** PC Gaming, Guitar, Podcasts, Reading (Books & Manga), Anime, Music, and Fitness.

EDUCATION

BMS Institute of Technology and Management, Bangalore, Karnataka Master of Computer Applications (MCA) Graduation CGPA: 8.47/10	February 2024 – February 2026
Rani Channamma University (A.S. Patil), Bijapur, Karnataka Bachelor of Computer Applications (BCA) Graduation CGPA: 8.73/10	September 2020 - October 2023
Roopadevi PU Science College, Bijapur, Karnataka Pre-University Course (PUC) in Science Percentage: 68%	June 2018 - July 2020
Prerana Public School (CBSE), Bijapur, Karnataka Secondary School Leaving Certificate (SSLC)	July 2018